Variables and Data types

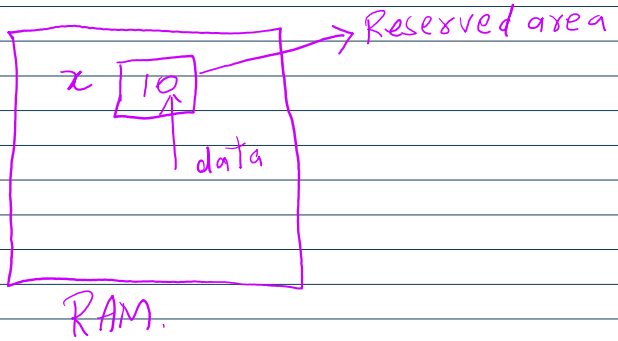
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What is a variable in Java?

In math it represents unknown quantity, or assumption. Same in programming also, but in programming it has extra property like it hold a memory, where we can store our value.

```
int x = 10;
```

A variable is a name of a reserved area allocated in memory.
In other words, it is a name of the memory location.
It is a combination of "vary + able", which means its value can be changed.

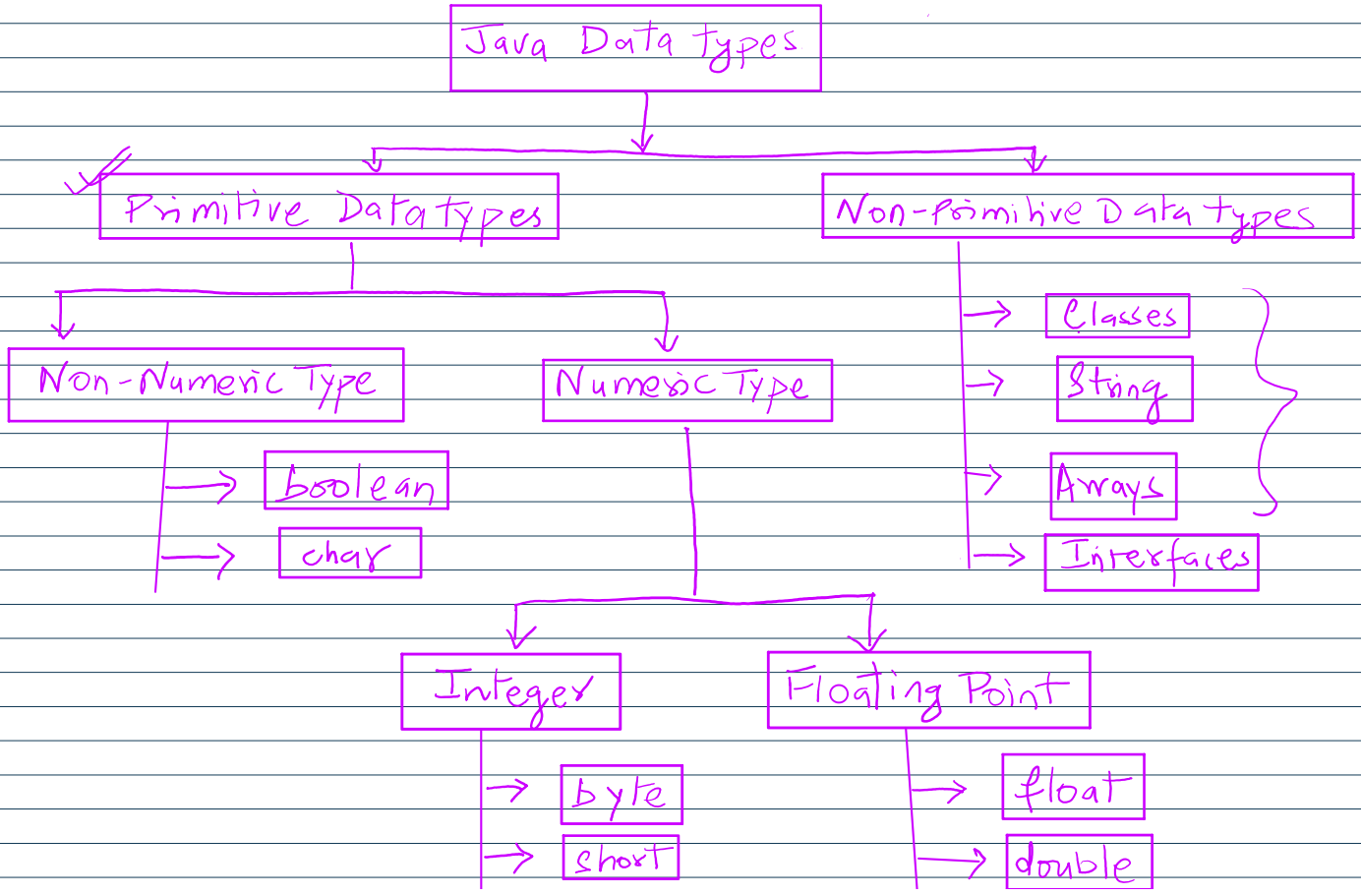


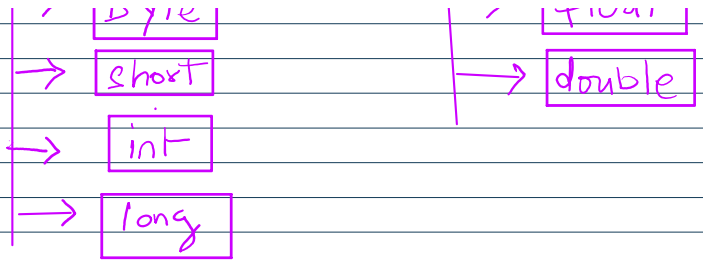
Data types in Java

In programming languages, data types specify the different sizes and values that can be stored in the variable or constants. Each data type is predefined, which makes Java a statically and strongly typed language. There are following two types of data types in Java.

1. **Primitive type:** The primitive data type include boolean, char, byte, short, int, long, float and double.

2. **Non-Primitive Data type:** The non primitive data type include classes, interfaces, string and arrays.





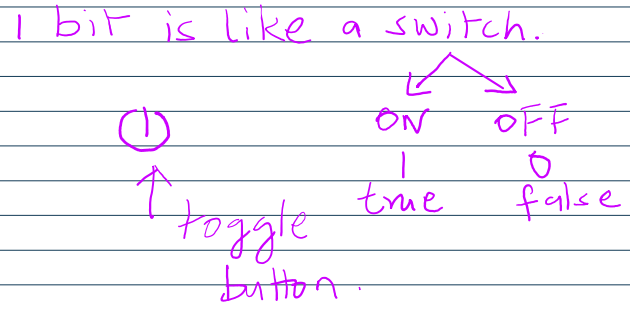
Java Primitive Data types:

1. **Boolean type:** In Java, the Boolean data type represents a single bit of information with two possible states, **true** or **false**. The size of boolean data type is one bit.

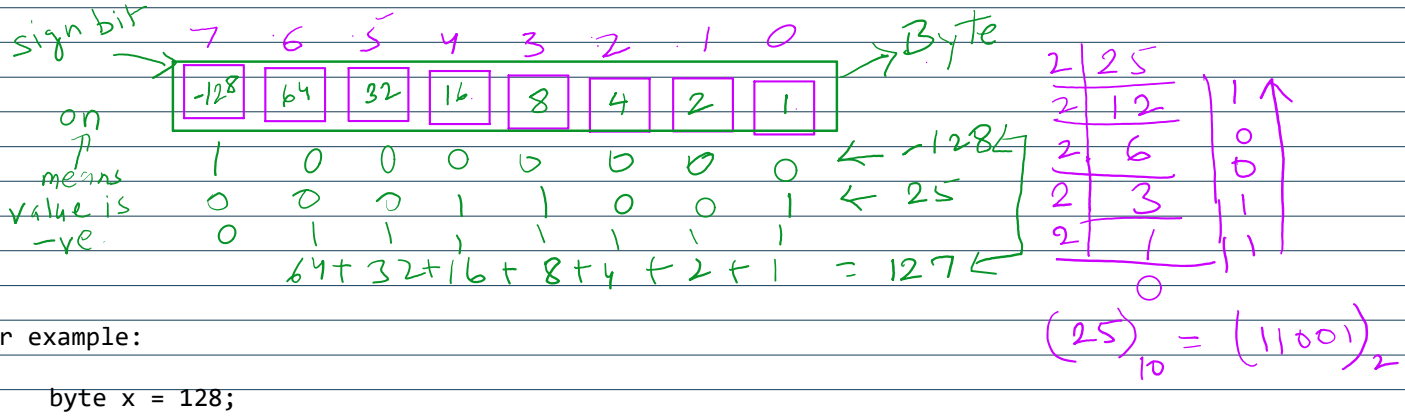
   It is used to store the result of logical expressions or conditions, unlike any other primitive data types, like int or double boolean does not have a specific size or range. It is typically implemented as a single bit, although exact implementation may vary across platforms.

   For example:

```
boolean b =true;
b
true   (boolean)
b = 5 > 6;
b
false   (boolean)
```



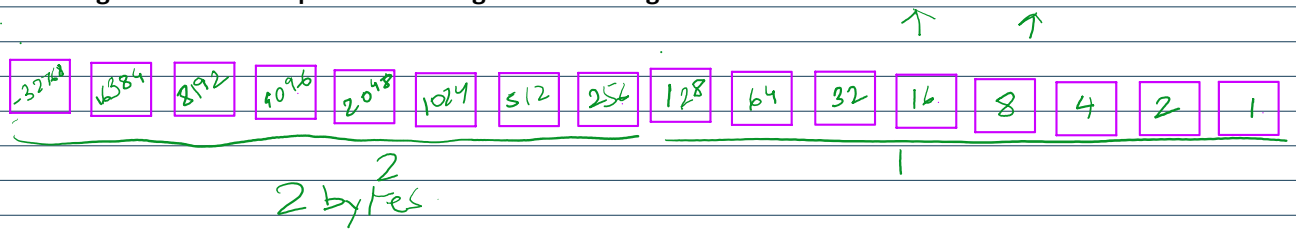
2. **"byte" Data type:** The bite data type in Java is a primitive data type that represents an 8 bit signed two's complement integer. It has a range of values from -128 to 127. Its default value is zero.



For example:

```
byte x = 128;
Error: incompatible types: possible lossy conversion from int to byte
byte x = 127;
x
127   (byte)
x = -128
-128   (byte)
x
-128   (byte)
x = -129;
Error: incompatible types: possible lossy conversion from int to byte
```

3. **"short" data type: The short data type in Java is a primitive data type that represents 16 bits signed two's complement integer. The range of value is -32768 to 32767.**



4. **"int" Data type:** The intact data type in Java is a primitive data type that represents a 32 bit signed two's complement integer. It has a range of values from -214783648 to 214783647.

5. **"long" Data type:** The long data type in Java is a primitive data type that represents a 64 bit signed two's complement integer. It has a wider range of values than int, ranging from nine to -23327036845775808 to 23327036845775807. It's default value is 0L or 0l.

For example:

```
long k = 0;
k
0   (long)
long m = 34546789992993;
Error: integer number too large
long m = 34546789992993L;
m
34546789992993   (long)
m = 345467899929931;
m
34546789992993   (long)
```

default numeric values in Java is: 1. int, 2. double

to use other types, like

1. float = f or F
2. long = l or L

6. **"float" Data type:** The float data type in Java is a primitive data type that represents single precision, 32-bits, IEEE 754 floating point numbers. It can represent a wide range of decimal values, but it is not suitable for precise values such as currency. It is default value is 0.0F or 0.0f.

32-bits = 4 bytes.

For Example:

```
float f = 0;
float g = 0.0;
Error: incompatible types: possible lossy conversion from double to float
float g = 0.0f;
float h = 0.0F;
```

↑ higher data type

It supports 6 decimal place digits. Meaning, after decimal point 6 digits.

7. **"double" Data Type:** The double data type in Java is a primitive data type that represents double precision 64 bits. IEE 754 floating point numbers. Its default value is 0.0. It provides a wide range of values and greater precision compared to the flow data type, which make it suitable for applications where accurate representation of decimal values is required.
It supports 15 decimal place digits. Meaning, after decimal point 15 digits.
For example:

```
double d = 0.0;
double q = 9.987654324;
```

```
q
9.987654324   (double)
d
0.0   (double)
```

8. **"char" Data type:** The car data type in Java is a primitive data type that represents single 16 bits UNICODE character. It can store any character from the Unicode character set, which allows Java to support the internationalization and representation of characters from various languages and writing systems.
   Unicode characters are preceded by special characters called "\uXXXX".

   For example:

```
char c = 'A';
c
'A'   (char)
char w = '@';
w
'@'   (char)
char t = '\u20B9";
Error: unclosed character literal
char t = '\u20B9';
t
'₹'   (char)
System.out.print("\n \u0915 \u0928 \u092d");
```

Data type	Default Values	Size	Range of values
boolean	false	1 bit	true or false
byte	0	8 bit or 1 byte	-128 to 127
char	'\u0000' or null	16 bits or 2 bytes	'\u0000' to '\uffff'
short	0	16 bits or 2 bytes	-32768 to 32767
int	0	32 bits or 4 bytes	-2147483648 to 2147483647
long	0	64 bits or 8 bytes	-9223372036854775808 to 9223372036854775808
float	0.0f or 0.0F	32 bits or 4 bytes	we will calculate later.
double	0.0d or 0.0D	64 bits or 8 bytes	we will calculate later.

Range of value of integer types:

Formulae: $-1(2)^{n-1}$ to 0 to $2^{n-1}-1$

$\because n$ = number of bits

byte = $-1(2)^{8-1}$ to 0 to $2^{8-1}-1$

-128 to 0 to 127